

Performance Analysis of Floating Solar Photovoltaic Systems for Humid and Saline Conditions: A Coupled Thermal, Soiling and Degradation Assessment

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Abstract—Floating photovoltaic (FPV) plants are usually credited with higher energy yield than land-based plants because the water body cools the modules. In hot, humid, coastal regions the same water body also supplies moisture and, at brackish or near-shore sites, airborne salt. Both accelerate soiling and degradation. This paper presents a coupled hourly model that captures these competing effects. The model combines an over-water microclimate, a humidity-dependent spectral correction, Faïman module temperature with dew formation, a dust and sea-salt soiling balance with NaCl deliquescence, and a four-mechanism degradation model: intrinsic, Peck temperature-humidity with lagged encapsulant moisture, potential-induced degradation (PID) with salt-assisted surface leakage, and chloride corrosion scaled by time of wetness. Three 1-MWp plants at Visakhapatnam, India, are compared: coastal land-mounted, freshwater FPV and saline FPV. Freshwater FPV runs 2.8 K cooler in daytime and yields 1636 kWh/kWp (+1.6%) against 1611 kWh/kWp on land. Saline FPV loses this advantage in the first year (-1.5%). Over 25 years the humid air over water cancels the Arrhenius benefit of cooler modules for temperature-humidity degradation. At the saline site, corrosion and PID raise total degradation to 1.24%/yr against 0.93%/yr on land, so lifetime energy falls by -4.9%. PID-resistant, IEC 61701-certified modules recover 4.2% of lifetime energy. The results hold over 20 independent weather realizations and are most sensitive to the salt load.

Index Terms—Floating photovoltaics, humidity, salinity, soiling, potential-induced degradation, corrosion, module temperature, energy yield, degradation rate, levelized cost of energy.

NOMENCLATURE

G	Plane-of-array (POA) irradiance (W/m^2).
T_a, T_d, T_m	Air, dew-point and module temperature ($^{\circ}\text{C}$).
v	Wind speed (m/s).
U_0, U_1	Faïman heat-loss coefficients.
RH_s, RH_e	Module-surface and encapsulant relative humidity (%).
m_d, m_s	Dust and salt surface density (g/m^2).
SR	Soiling ratio.
TOW	Time of wetness (h/yr).
D_{Cl}	Chloride deposition rate ($\text{mg}/(\text{m}^2 \text{ d})$).

R	Annual power degradation rate (%/yr).
E_a	Activation energy (eV).
k_B	Boltzmann constant ($8.617 \times 10^{-5} \text{ eV/K}$).
PR	Performance ratio.

I. INTRODUCTION

FLOATING photovoltaic (FPV) systems place PV arrays on pontoons over reservoirs, lakes, lagoons and sheltered coastal waters. They avoid competition for land, reduce evaporation from water-supply reservoirs and can share grid infrastructure with hydropower plants [1]–[4]. India has a large inventory of reservoirs, salt pans and backwaters along a coast with high irradiation. FPV is therefore attractive there, but many candidate sites are hot, humid and exposed to marine aerosol.

The performance case for FPV usually rests on the cooling effect of the water. Field studies in tropical Singapore found lower module temperatures and higher performance ratios for well-ventilated FPV designs than for a rooftop reference [5]. Comparative measurements in the Netherlands and Singapore showed that the effective heat-loss coefficient of FPV depends strongly on how open the floating structure is [6]. Simulation of offshore systems likewise predicts higher yield than for land-based systems, mainly because the modules run cooler [7]. Most of these assessments focus on first-year energy [8].

The cooling, however, comes with more humid air. Humidity and temperature together drive hydrolysis, delamination and corrosion in PV modules. The Peck model describes this damage as an Arrhenius function of temperature multiplied by a power of relative humidity [9], and moisture ingress through polymer backsheets takes hours to days [10]. Combined-stressor lifetime models confirm that humidity is a first-order variable for outdoor degradation [11]. Potential-induced degradation (PID) is strongly accelerated by humidity and surface conductivity [12]–[14]. Field surveys report higher degradation rates in hot climates than in temperate ones [15]; in India the hot and humid zones are among the worst affected [16]. At saline sites, chloride deposition and long time of wetness define the atmospheric corrosivity class of the site [17]. Salt mist is a recognized qualification stress for PV modules [18]. Sea-salt deposits also add to soiling, which is a major loss in sunny climates [19], [20].

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The simulation code, input parameters and all scripts used to produce the results are openly available (see the Data Availability section).

What is missing is an analysis that couples these mechanisms through the same microclimate. If the water body lowers module temperature but raises humidity, the net effect on lifetime energy is not obvious. It also differs between freshwater and saline sites. This paper addresses that gap with the following contributions:

- 1) A coupled hourly model chain in which one over-water microclimate drives module temperature, dew, spectral response, soiling and four degradation mechanisms, so their trade-offs are resolved consistently.
- 2) A moisture formulation based on module-surface relative humidity with a first-order encapsulant lag. It captures why cooler modules see more moisture, and it couples this to PID through salt-assisted surface leakage.
- 3) A quantitative comparison of coastal land PV, freshwater FPV and saline FPV for a representative hot-humid Indian site. The comparison covers first-year losses, mechanism-resolved degradation, 25-year energy and levelized cost of energy (LCOE), with and without PID- and salt-mist-resistant modules.
- 4) A humidity-salinity sensitivity map and a 20-realization weather ensemble that separate robust conclusions from weather noise.

The rest of the paper is organized as follows. Section II describes the site and the scenarios. Section III presents the model. Section IV reports the results, Section V discusses implications and limitations, and Section VI concludes.

II. SITE AND STUDY SCENARIOS

The reference site is Visakhapatnam (17.69° N, 83.22° E) on the Bay of Bengal. Its climate is tropical, with a hot pre-monsoon season (April–May), the south-west monsoon (June–September) and the north-east monsoon with cyclonic activity (October–November). An hourly typical year was synthesized from monthly normals (Section III-A). It has a global horizontal irradiation (GHI) of 1,889 kWh/m², a mean air temperature of 28.1 °C, a mean relative humidity of 76.4% and 1,043 mm of rain (Fig. 1).

Three 1-MWp (DC) plants with a DC/AC ratio of 1.2 and monocrystalline PERC modules (−0.35 %/K) are compared. Their parameters are listed in Table I:

- *Land PV*: a coastal ground-mounted plant on open racks with 15° tilt (the reference).
- *FPV freshwater*: a pontoon plant with 12° tilt on an inland reservoir with low chloride deposition.
- *FPV saline*: the same pontoon design on near-shore or brackish water (lagoon, salt pan or harbour) with high salt and chloride loading.

Each plant is also simulated with PID-resistant modules certified to IEC 61701 severity level 6 (“mitigated”, M).

III. MODELING FRAMEWORK

The model chain is shown in Fig. 2. All sub-models run at hourly resolution over one year. Degradation rates are computed from the hourly stresses and then projected over the 25-year life of the plant. The established irradiance and electrical models follow their reference formulations, as implemented, for example, in pvlib [21].

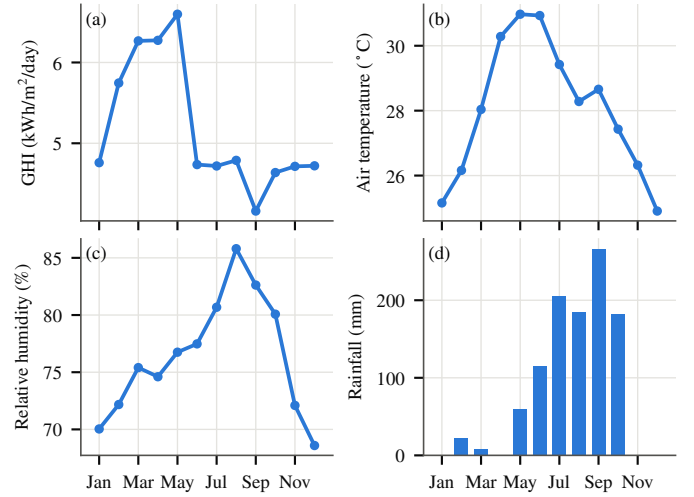


Fig. 1. Monthly climate of the synthetic Visakhapatnam typical year: (a) daily GHI, (b) air temperature, (c) relative humidity and (d) rainfall.

TABLE I
SCENARIO PARAMETERS

Parameter	Land	FPV fresh	FPV saline
Tilt (°)	15	12	12
Albedo	0.20	0.07	0.07
U_0 (W/(m ² K))	25.0	35.0	35.0
U_1 (W s/(m ³ K))	6.84	8.0	8.0
Air–water coupling α	0	0.25	0.30
Dew-point rise ΔT_d (K)	0	1.0	1.5
Wind factor k_v	1.00	1.15	1.25
Dust deposition (g/(m ² d))	0.15	0.04	0.03
Salt deposition (g/(m ² d))	0.015	0.010	0.060
D_{Cl} (mg/(m ² d))	30	8	180
Wiring loss (%)	1.5	2.0	2.0
Wave-motion mismatch (%)	0	0.7	1.0
Availability (%)	99.0	98.5	98.0
CAPEX (INR/Wp)	35	45	52
O&M (INR/kWp/yr)	450	550	750

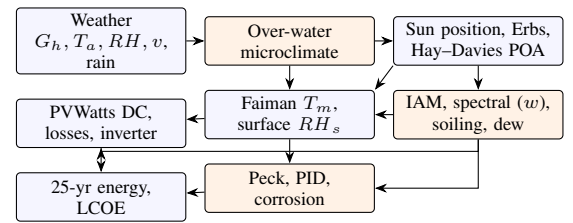


Fig. 2. Coupled model chain. Shaded blocks carry an explicit dependence on humidity or salinity.

A. Weather and Over-Water Microclimate

The synthetic year draws daily rain occurrence and amounts, a first-order autoregressive daily clearness index and daily temperature anomalies from monthly normals. Hourly GHI is the Haurwitz clear-sky value [22] scaled by the clearness index. Air temperature follows an asymmetric diurnal cycle, with the minimum near 05:00 and the maximum near 14:00. Holding the dew point T_d constant over each day reproduces the observed diurnal swing in relative humidity. Wind speed includes an afternoon sea breeze. Measured series can be used in place of the synthetic year.

Over water, the air temperature is pulled towards the water temperature T_w , taken as the 30-day running mean of air temperature plus 0.5 K. The dew point rises by ΔT_d because of evaporation, and the wind speed increases over the smoother surface:

$$\begin{aligned} T'_a &= T_a + \alpha (T_w - T_a), & v' &= k_v v, \\ T'_d &= \min(T_d + \Delta T_d, T'_a). \end{aligned} \quad (1)$$

Relative humidity follows from the Magnus relation $e_s(T) = 6.112 \exp[17.62T/(243.12 + T)]$ hPa as $RH = 100 e_s(T'_d)/e_s(T'_a)$.

B. Irradiance, Incidence and Spectrum

Solar position uses Spencer's series [23]. GHI is decomposed with the Erbs correlation [24] and transposed to the array plane with the Hay–Davies model [25]:

$$\begin{aligned} G &= G_b \cos \theta + G_d \left[A_i R_b + (1 - A_i) \frac{1 + \cos \beta}{2} \right] \\ &\quad + \rho G_h \frac{1 - \cos \beta}{2}, \end{aligned} \quad (2)$$

where θ is the angle of incidence, β the tilt, A_i the anisotropy index, R_b the beam ratio and ρ the albedo. Water has a much lower albedo than land (0.07 against 0.20). Beam reflection losses use the ASHRAE modifier $IAM = 1 - b_0(1/\cos \theta - 1)$ with $b_0 = 0.05$; diffuse and ground-reflected light take a constant factor of 0.95.

Humid air absorbs mainly in infrared bands where crystalline silicon responds weakly. The spectral mismatch factor for mono-Si is therefore computed from the Kasten–Young air mass AM [26] and the precipitable water w (cm), estimated from surface temperature and humidity [27]:

$$M = c_1 + c_2 AM + c_3 w + c_4 \sqrt{AM} + c_5 \sqrt{w} + c_6 AM / \sqrt{w}, \quad (3)$$

with the mono-Si coefficients of [28].

C. Module Temperature, Surface Humidity and Dew

Module temperature follows the Faïman model [29]. A net long-wave loss of $q_{\text{sky}} = 40 \text{ W/m}^2$ to the night sky applies when $G < 50 \text{ W/m}^2$:

$$T_m = T'_a + \frac{G - q_{\text{sky}} \mathbb{I}[G < 50]}{U_0 + U_1 v'}. \quad (4)$$

The land coefficients are the open-rack defaults. The FPV values lie within the range of heat-loss coefficients reported for pontoon systems [5], [6]. The relative humidity of the air layer in contact with the glass is

$$RH_s = 100 e_s(T'_d)/e_s(T_m), \quad (5)$$

so a hot module is effectively dry and a module colder than the dew point is wet. Dew forms when $T_m - T'_d < 1 \text{ K}$ and persists after sunrise until $T_m - T'_d > 4 \text{ K}$. While dew is present, a 4% optical loss applies.

D. Dust and Sea-Salt Soiling

Dust and salt surface densities evolve by a daily mass balance. Dust accumulates at r_d , and salt at $r_s v/v_0$ with $v_0 = 3 \text{ m/s}$, because sea-spray generation rises with wind. Rain above 2 mm/day removes a fraction $\eta_d f_\beta$ of the dust and $\eta_s f_\beta$ of the salt. Here $\eta_d = 0.80$, $\eta_s = 0.95$ (salt is soluble) and $f_\beta = \sin \beta / \sin 20^\circ$, so cleaning is less effective at low tilt. When the daily maximum RH exceeds the NaCl deliquescence point of 75%, the dissolved salt recrystallizes as a crust that cements dust to the glass. On such days the dust-cleaning efficiency is halved. All plants are cleaned every 15 days with 95% efficiency. The soiling ratio is

$$SR = \max(SR_{\min}, 1 - k_d m_d - k_s m_s), \quad (6)$$

with $k_d = 0.035$ and $k_s = 0.060 \text{ m}^2/\text{g}$.

E. Electrical Model

The effective irradiance G_{eff} includes the incidence, spectral, soiling and dew corrections. DC power follows the PVWatts model [30]:

$$P_{\text{dc}} = P_0 \frac{G_{\text{eff}}}{1000} [1 + \gamma(T_m - 25)] \prod_j (1 - \ell_j), \quad (7)$$

where $\gamma = -0.0035/\text{K}$ and ℓ_j are the wiring, mismatch, wave-motion and light-induced degradation (LID, 1%) losses. The PVWatts inverter curve (nominal efficiency 98%) applies, with clipping at the AC rating, followed by the availability factor. Each loss is reported as a percentage of the energy at the preceding stage of the chain.

F. Degradation Model

The annual degradation rate is the sum of four mechanisms:

$$R = R_0 + R_{\text{TH}} + R_{\text{PID}} + R_{\text{C}}. \quad (8)$$

The intrinsic term $R_0 = 0.30\%/yr$ covers ultraviolet exposure and thermal cycling and does not depend on the site.

Temperature–humidity. Moisture inside the encapsulant follows the surface humidity with a first-order lag of time constant $\tau = 48 \text{ h}$, representing diffusion through the back-sheet [10]:

$$RH_e(t) = RH_e(t-1) + \kappa [RH_s(t) - RH_e(t-1)], \quad (9)$$

with $\kappa = 1 - e^{-\Delta t/\tau}$. With the Arrhenius factor

$$a_E(T_m) = \exp\left[-\frac{E}{k_B} \left(\frac{1}{T_m} - \frac{1}{T_{\text{ref}}}\right)\right], \quad (10)$$

the Peck rate [9] is averaged over the year:

$$R_{\text{TH}} = R_{\text{TH}}^{\text{ref}} \left\langle a_{E_a}(T_m) \left(\frac{RH_e}{RH_{\text{ref}}}\right)^n \right\rangle, \quad (11)$$

with $E_a = 0.79 \text{ eV}$, $n = 3$, $T_{\text{ref}} = 25^\circ\text{C}$ and $RH_{\text{ref}} = 60\%$. Because RH_s depends on T_m , cooling a module lowers the Arrhenius factor but raises RH_e ; (11) resolves this trade-off.

Potential-induced degradation. PID acts only while the array is energized ($G > 50 \text{ W/m}^2$). Its rate depends on

module temperature and on the probability that a conductive film covers the glass:

$$R_{\text{PID}} = R_{\text{PID}}^{\text{ref}}(1 - \mu_{\text{PID}})\langle a_{E_{\text{PID}}}(T_m) f_{\text{leak}} \rangle_{\text{on}}, \quad (12)$$

$$f_{\text{leak}} = \left[1 + \exp \left(-\frac{RH - (RH_{50} - \Delta_s s)}{5} \right) \right]^{-1}, \quad (13)$$

where $E_{\text{PID}} = 0.9 \text{ eV}$, $RH_{50} = 70\%$, and $s = \min(m_s/0.5, 1)$ is the salt coverage. A full hygroscopic salt film lowers the threshold by $\Delta_s = 15\%$ RH. The factor $\mu_{\text{PID}} = 0.9$ applies to PID-resistant modules [12]–[14].

Chloride corrosion. Following the ISO 9223 logic of time of wetness and chloride deposition [17], the corrosion rate is

$$R_C = R_C^{\text{ref}} \frac{TOW}{TOW_{\text{ref}}} \sqrt{\frac{D_{\text{Cl}}}{D_{\text{ref}}}} (1 - \mu_C), \quad (14)$$

where TOW counts the hours with $RH > 80\%$ or with dew on the module. The reference values are $TOW_{\text{ref}} = 2500 \text{ h}$ and $D_{\text{ref}} = 60 \text{ mg}/(\text{m}^2 \text{ d})$, and $\mu_C = 0.5$ for IEC 61701 severity-6 modules [18].

The reference rates ($R_{\text{TH}}^{\text{ref}} = 0.10$, $R_{\text{PID}}^{\text{ref}} = 0.06$ and $R_C^{\text{ref}} = 0.10\%/ \text{yr}$) were calibrated so that standard modules on land at this site degrade by roughly $1\%/ \text{yr}$. That is the order of magnitude reported for hot-humid Indian climates [16], and above the global median of about $0.5\%/ \text{yr}$ [15].

G. Lifetime Energy and Economics

Year- y energy is $E_y = E_1(1 - R/100)^{y-1}$ over a life of $N = 25$ years. The LCOE is

$$\text{LCOE} = \frac{C_0 + \sum_{y=1}^N O_y(1+r)^{-y}}{\sum_{y=1}^N E_y(1+r)^{-y}}, \quad (15)$$

with a discount rate $r = 9\%$ and O&M costs O_y that escalate by $5\%/ \text{yr}$. The costs in Table I are illustrative Indian-market values. For the freshwater plant, the evaporation avoided by shading ($10 \text{ m}^2/\text{kWp}$, 65% suppression of $1800 \text{ mm}/ \text{yr}$ pan evaporation) is reported as a co-benefit.

IV. RESULTS

A. Thermal Behavior

Fig. 3 shows the mean diurnal module temperature in May, the hottest month. Over the year, floating modules run 2.8 K cooler in daytime than land-mounted modules, and their peak temperature is 6.9 K lower (Table II). The mean daytime module-to-air temperature rise falls from 9.5 K on land to 6.9 K (freshwater) and 6.6 K (saline). The saline plant runs slightly cooler still because of its stronger air–water coupling and higher wind. At night the picture reverses. Modules on land radiate below the air temperature, while over water the air stays warmer and the modules follow it; yet the higher dew point keeps floating modules closer to condensation. Dew is present on the glass for $1,533 \text{ h}$ on land, $1,615 \text{ h}$ on freshwater and $1,879 \text{ h}$ on saline water.

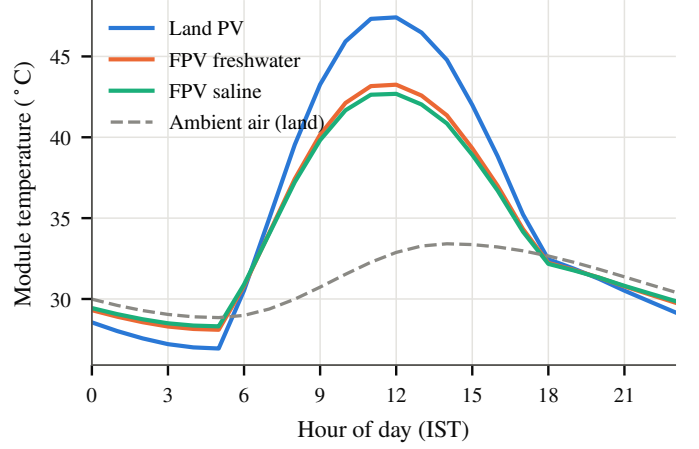


Fig. 3. Mean module temperature by hour of day in May for the three plants, with the site air temperature.

TABLE II
YEAR-1 PERFORMANCE

Metric	Land	FPV fresh	FPV saline
POA irradiation ($\text{kW h}/\text{m}^2$)	1963	1953	1953
Specific yield (kWh/kWp)	1611	1636	1587
Performance ratio	0.821	0.838	0.813
CUF, DC basis (%)	18.39	18.68	18.12
Mean daytime T_m ($^{\circ}\text{C}$)	38.8	36.1	35.8
Maximum T_m ($^{\circ}\text{C}$)	56.1	49.2	48.6
Mean $T_m - T_a$, daytime (K)	9.5	6.9	6.6
Mean air RH at array (%)	76.4	80.0	82.1
Dew on glass (h/yr)	1,533	1,615	1,879
Mean daytime soiling ratio	0.964	0.987	0.965

B. Energy Yield and Loss Breakdown

Freshwater FPV gives the highest first-year specific yield, $1636 \text{ kWh}/\text{kWp}$, which is 1.6% above the land plant. Its PR is 0.838 against 0.821 (Table II). The loss breakdown in Table III and Fig. 4 shows where the gain comes from. The temperature loss falls from 5.84% to 4.58% , and the soiling loss falls from 3.76% to 1.34% because little dust reaches the water surface. Part of this gain is given back by DC losses (longer cable runs and wave-induced mismatch: 4.62% against 3.46%), by lower availability and by slightly lower POA irradiation from the lower tilt and the low albedo of water.

The saline plant has the same thermal advantage (4.44%). Salt soiling, however, brings its soiling loss back to the land level (3.67%). With the highest DC and availability losses, its first-year yield is $1587 \text{ kWh}/\text{kWp}$ (-1.5% against land). Humidity itself has a small but beneficial spectral effect: precipitable water shifts the spectrum towards wavelengths that c-Si converts well, which gives a gain of about 0.25% at all three sites. Dew causes a loss of less than 0.1% because condensate evaporates within the first hours after sunrise.

Fig. 5 shows the monthly PR. The FPV advantage is largest in the dry, clear months (November–May). In the pre-monsoon months the saline plant performs worst: heavy salt deposition under the strong onshore wind meets a lack of rain. During the monsoon frequent rain cleans all three arrays and the thermal advantage shrinks under cloudy skies, so the gap narrows.

TABLE III
YEAR-1 LOSS BREAKDOWN (% OF PRECEDING STAGE)

Loss stage	Land	FPV fresh	FPV saline
Incidence angle (IAM)	3.44	3.47	3.47
Spectral (humidity)	-0.24	-0.24	-0.25
Soiling (dust + salt)	3.76	1.34	3.67
Dew / condensation	0.04	0.07	0.08
Module temperature	5.84	4.58	4.44
DC (wiring, mismatch, motion, LID)	3.46	4.62	4.91
Inverter and clipping	2.03	2.03	2.04
Availability	1.00	1.50	2.00

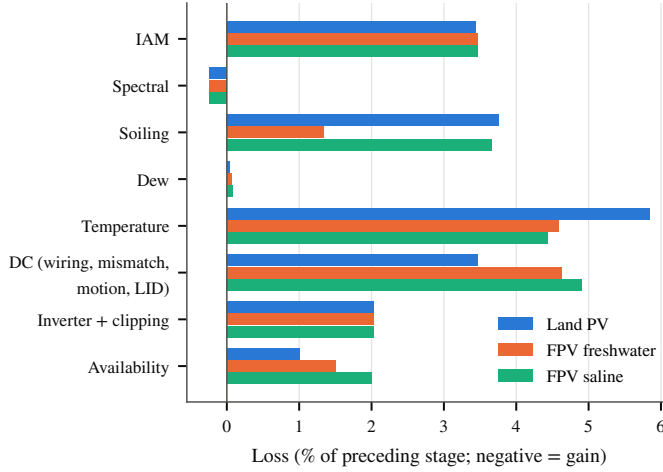


Fig. 4. Year-1 loss breakdown. Negative values are gains.

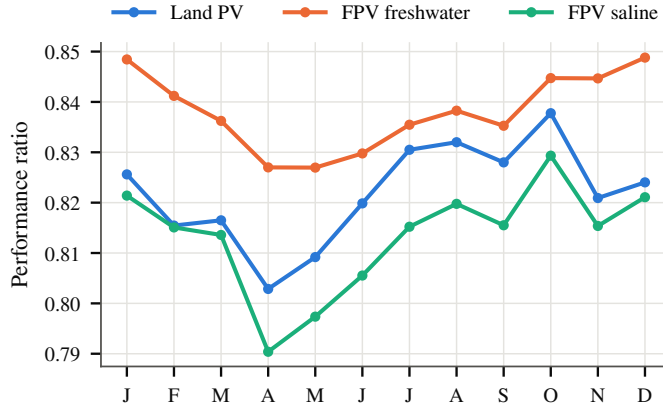


Fig. 5. Monthly performance ratio.

Fig. 6 shows the daily soiling ratio. Between cleanings, salt deposition on the saline plant produces a saw-tooth pattern similar to dust soiling on land, while the freshwater array stays close to clean.

C. Degradation

Table IV and Fig. 7 resolve the degradation rate by mechanism. The central result is the temperature–humidity term. Floating modules are cooler, but their encapsulant equilibrates at a higher moisture level: mean RH_e is 65.8% on land, 70.9% on freshwater and 72.9% on saline water. Through the cubic humidity dependence of (11), this cancels the Arrhenius

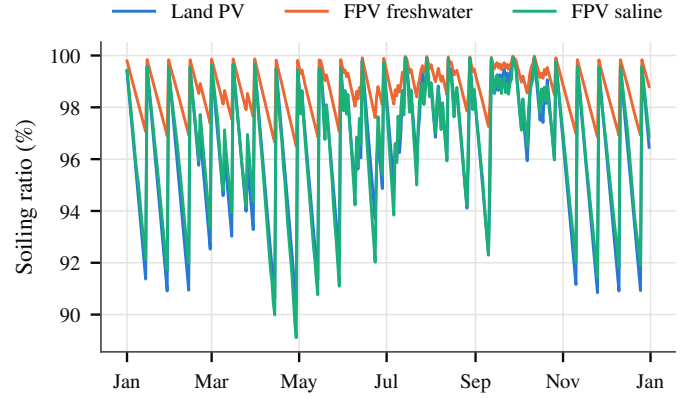


Fig. 6. Daily soiling ratio with rain cleaning, NaCl deliquescence and 15-day manual cleaning.

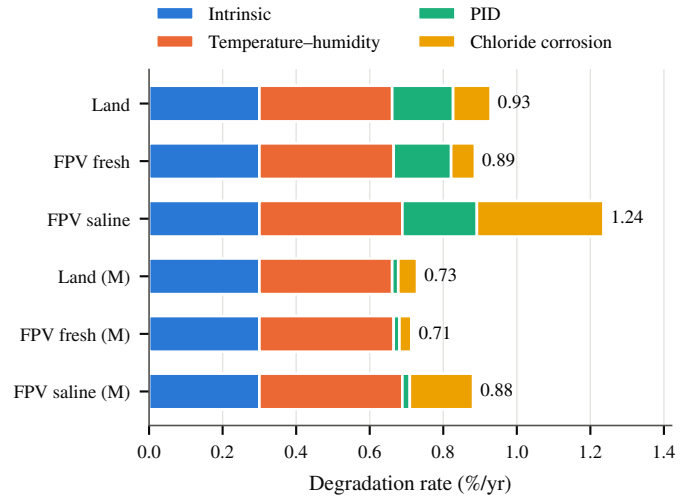


Fig. 7. Annual degradation rate by mechanism. (M): PID-resistant, IEC 61701 severity-6 modules.

benefit. R_{TH} is 0.36%/yr on land, 0.37%/yr on freshwater and 0.39%/yr on saline water. For hydrolysis-type damage, *cooling by water does not translate into slower ageing at a humid site.*

PID is slightly lower on freshwater (0.16%/yr) than on land (0.17%/yr), because the higher activation energy rewards cooler modules. On saline water it is highest (0.20%/yr), because the hygroscopic salt film makes the glass conductive at lower RH. Corrosion separates the sites most clearly. The time of wetness grows from 3,609 h/yr on land to 4,432 h/yr and 4,967 h/yr over water. Combined with 180 mg/(m² d) of chloride deposition, this gives R_C = 0.34%/yr on saline water against 0.10%/yr on land and 0.06%/yr on freshwater. In total, the freshwater plant degrades at 0.89%/yr, close to the land plant (0.93%/yr), while the saline plant degrades at 1.24%/yr.

D. Lifetime Energy, Mitigation and Cost

Over 25 years (Fig. 8), freshwater FPV delivers 36,838 MWh/MWp, 2.1% more than the land plant. The saline plant delivers 34,313 MWh/MWp, which is -4.9% less, and keeps only 74.2% of its initial capacity in year 25. PID-resistant, IEC 61701 severity-6 modules reduce the saline

TABLE IV
DEGRADATION, LIFETIME ENERGY AND LCOE (PER MWp; M = PID-RESISTANT, IEC 61701 SEVERITY-6 MODULES)

	Land	FPV fresh	FPV saline	Land (M)	FPV fresh (M)	FPV saline (M)
R_0 intrinsic (%/yr)	0.30	0.30	0.30	0.30	0.30	0.30
R_{TH} temperature-humidity (%/yr)	0.36	0.37	0.39	0.36	0.37	0.39
R_{PID} (%/yr)	0.17	0.16	0.20	0.02	0.02	0.02
R_C chloride corrosion (%/yr)	0.10	0.06	0.34	0.05	0.03	0.17
Total R (%/yr)	0.93	0.89	1.24	0.73	0.71	0.88
Time of wetness (h/yr)	3,609	4,432	4,967	3,609	4,432	4,967
Mean encapsulant RH (%)	65.8	70.9	72.9	65.8	70.9	72.9
Capacity in year 25 (%)	79.9	80.8	74.2	83.9	84.2	80.9
25-year energy (MWh)	36,093	36,838	34,313	36,946	37,590	35,750
LCOE (INR/kWh)	2.84	3.55	4.47	2.90	3.61	4.45

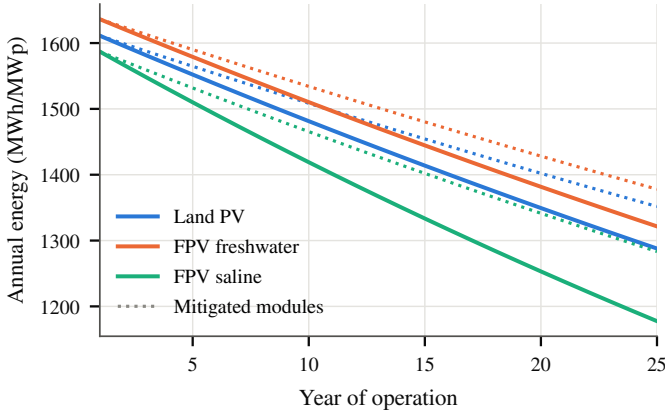


Fig. 8. Annual energy over 25 years of operation. Dotted lines: PID-resistant, IEC 61701 severity-6 modules.

degradation rate to 0.88%/yr and raise its lifetime energy by 4.2%. Even so, the mitigated saline plant does not reach the unmitigated land plant. On land and freshwater the same modules lower the rate to 0.73% and 0.71%/yr.

With the assumed costs, FPV is more expensive per kWh: 3.55 INR/kWh on freshwater and 4.47 INR/kWh on saline water, against 2.84 INR/kWh on land. The cost of the floats, mooring and marine-grade balance of system dominates over the yield gain. For the saline plant, the mitigated modules (assumed to cost 1.5 INR/Wp more) already lower the LCOE from 4.47 to 4.45 INR/kWh. FPV is therefore justified mainly by land savings and, on freshwater, by co-benefits: the freshwater plant avoids about 11,700 m³ of reservoir evaporation per MWp each year.

E. Sensitivity to Humidity and Salinity

For the saline plant, Fig. 9 varies the dew-point rise over water from 0 to 2.5 K and scales the salt and chloride deposition together from 0.25 $\times$ to 2 $\times$ the baseline. The degradation rate ranges from 0.84 to 1.57%/yr, and 25-year energy ranges from 32,002 to 36,759 MWh/MWp. Each kelvin of extra dew point adds about 0.158%/yr of degradation at baseline salinity, because the time of wetness grows from 3,030 to 6,169 h/yr across the sweep. Across the full ranges swept, salinity moves lifetime energy by about 3,191 MWh/MWp and moisture by about 1,580 MWh/MWp. Salinity also cuts first-year yield by 5.3% between the lowest and highest salt loads, through

TABLE V
DIFFERENCES FROM LAND PV OVER 20 WEATHER REALIZATIONS (MEAN $\pm$ SD)

	FPV fresh	FPV saline
Year-1 yield (%)	+1.63 $\pm$ 0.08	-1.49 $\pm$ 0.07
25-year energy (%)	+2.15 $\pm$ 0.09	-4.98 $\pm$ 0.08
Degradation rate (%/yr)	-0.043 $\pm$ 0.003	+0.312 $\pm$ 0.005

soiling. At the baseline moisture level (+1.5 K), the saline plant matches the lifetime energy of the land plant only if the salt load falls to $<0.25\times$ the baseline.

F. Robustness to Weather Variability

To check that the conclusions do not depend on one synthetic year, the three plants were simulated for 20 independent weather realizations (Table V, Fig. 10). The land yield varies between 1,573 and 1,652 kWh/kWp, but the differences between plants are stable. Freshwater FPV outperforms land in 20 of 20 years, and the standard deviations of the relative differences are below 0.1 percentage point. The ranking of the plants is the same in every realization.

V. DISCUSSION

A. Implications for Design and Siting

The results separate two questions that are often merged: whether FPV produces more energy in its first year, and whether it produces more over its life. For freshwater reservoirs in a hot-humid climate both answers are yes, by 1.6% and 2.1%, respectively. The lifetime gain comes from the thermal and soiling benefits, not from slower degradation. For saline sites both answers are no. A module bill of materials chosen for the environment narrows the lifetime gap but does not close it. These findings suggest the following:

- At saline or near-shore sites, specify PID-resistant modules certified to IEC 61701 severity 6. Glass-glass modules with low-permeability encapsulants reduce the encapsulant moisture that drives R_{TH} .
- Use negative-pole grounding or night-time PID recovery, marine-grade IP68 connectors and corrosion-resistant structures designed to ISO 9223 class C5/CX.
- At saline sites, shorten the cleaning interval in the dry, windy pre-monsoon season, when rain does not remove

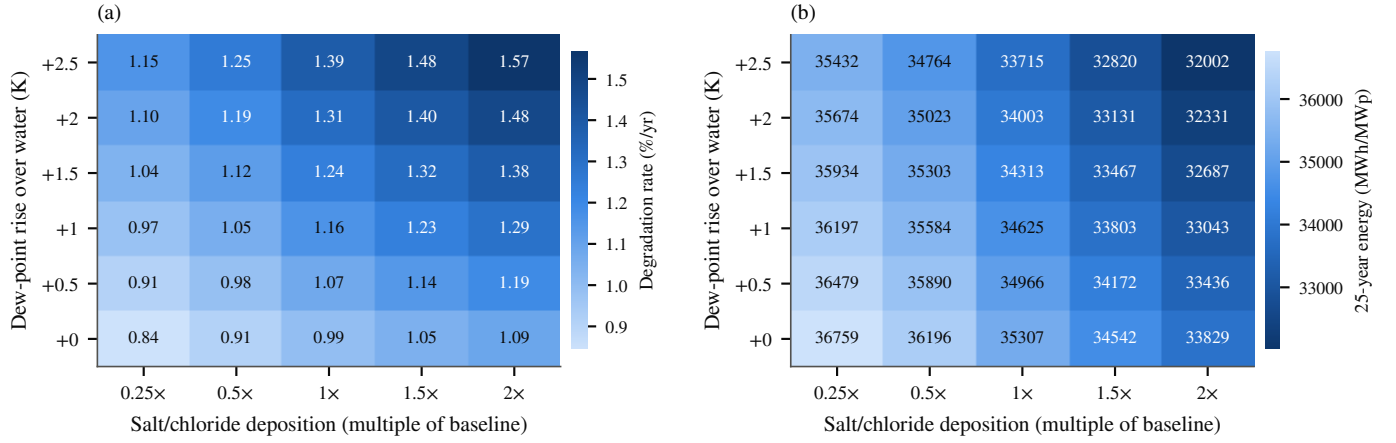


Fig. 9. Sensitivity of the saline FPV plant to the dew-point rise over water and the salt/chloride load: (a) total degradation rate and (b) 25-year energy.

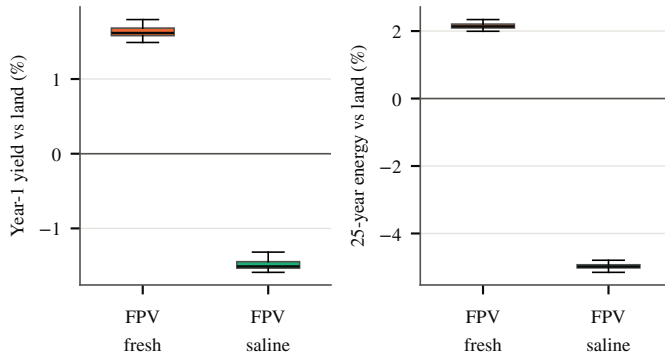


Fig. 10. Distribution over 20 weather realizations of the year-1 yield and 25-year energy of the FPV plants relative to land PV.

salt crusts. Clean with demineralized water to avoid re-depositing salt.

- Monitor module temperature, RH at the array, insulation resistance and soiling with reference cells, in line with IEC 61724-1 [31], so that the degradation parameters can be recalibrated with site data.

B. Comparison With Previous Work

The simulated thermal advantage of FPV, a daytime module-to-air rise lower by about 3 K and peak temperatures lower by about 7 K, agrees qualitatively with tropical field measurements [5], [6]. Those studies also show that the size of the effect depends strongly on how ventilated the float design is. The first-year gains of a few percent are consistent with simulation studies that attribute the FPV advantage mainly to cooling [7]. The novelty here is the explicit counter-effect of humidity on degradation. A first-year comparison alone would miss it.

C. Limitations

The weather is a synthetic typical year built from monthly normals, not a measured year. The over-water microclimate parameters (α , ΔT_d , k_v), the deposition rates and the degradation reference rates are engineering assumptions informed by

the literature, not site measurements. Absolute values should therefore be read as indicative, whereas the differences between plants, which share all common assumptions, are more robust (Table V). The degradation model is linear in time and additive across mechanisms; it does not capture interactions such as the saturation of PID or the abrupt failure of corroded interconnects. Structural fatigue of floats and moorings, bird soiling, bio-fouling and storm damage are not modeled. The framework accepts measured hourly weather, and future work will calibrate it against monitoring data from operating FPV plants in coastal Andhra Pradesh.

VI. CONCLUSION

A coupled thermal, soiling and degradation model was used to compare land-mounted, freshwater floating and saline floating PV for a hot, humid, coastal site in India. Water cooling lowers daytime module temperature by 2.8 K and gives freshwater FPV the highest first-year yield (+1.6%) and lifetime energy (+2.1%). The same water body raises the humidity at the modules. This cancels the expected temperature benefit for moisture-driven degradation, and at saline sites it combines with chloride deposition to make corrosion and PID the decisive mechanisms. Saline FPV then degrades at 1.24%/yr and loses -4.9% of lifetime energy relative to land. Module selection for PID and salt-mist resistance is the most effective mitigation. Salinity is a stronger driver than the extra moisture, so site selection should give it priority. Assessments of FPV in humid and saline regions should therefore report lifetime energy with mechanism-resolved degradation, not first-year yield alone.

DATA AVAILABILITY

The open-source Python implementation (`fpv_analysis`), the scenario parameters, the generated result tables and the script that produces every number and figure in this paper (`paper/make_paper_assets.py`) are available in the project repository <https://github.com/subrahmanyamtanala-rgb/floating-solar-photovoltaic-system-for-humid-and-saline-conditions>.

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